



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

LSVTE302

SURVEYING TOOLS AND EQUIPMENT

Identify surveying tools and equipment

Competence

RQF Level: 3

Learning Hours

Credits: 3



Sector: Construction and Building Services

Trade: Land Surveying

Module Type: General

Curriculum: CBSLSV3001- TVET Certificate 3 in Land Surveying

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Purpose statement	This core module describes the skills, knowledge and attitudes required for the learners to carry out the identification and selection of surveying instruments and tools with respect to the concerned work.					
Leaning assumed to be in place	N/A					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	100%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment		N.A.	

Elements of Competency and Performance criteria

Elements of competency	Performance criteria
1. Identify distance measuring tools and instruments.	1.1. Distance measuring tools are correctly identified according to their types.
	1.2. Distance measuring instruments are properly identified in accordance with their area of application.
	1.3. Distance measuring tools and instruments are clearly described according to their functionalities.
2. Identify levelling tools and instruments.	2.1. Levelling tools are clearly identified according to their types.
	2.2. Different levelling instruments are properly identified according to their types.
	2.3. Levelling instruments and tools are properly described according to their functionalities.
3. Identify angular measuring tools and instruments.	3.1. Angular measuring instruments are identified in accordance with their functionalities.
	3.2. Angular measuring tools are clearly identified according to their types.

	3.3. Angular measuring instruments and tools are well described according to their functionalities.
4. Identify coordinate tracking instruments.	4.1 Coordinate tracking instruments are properly identified.
	4.2 Coordinate satellite-based tracking instruments are properly described based on their functionalities and accuracy.
	4.3 Coordinate ground-based tracking instruments are clearly described based on their functionalities and accuracy.

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Identify distance measuring tools and instruments. 2. Identify levelling tools and instruments. 3. Identify angular measuring tools and instruments. 4. Describe coordinate tracking instruments.
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Identify distance measuring tools and instruments	Learning hours:8
Indicative content	
* Surveying tools used in distance measurement, (Refer to ISO 8322-4, ISO 17123-2) Calibrated Tape measure, chain, measuring wheel, distometer, * Types of surveying instruments used in distance measurement • Direct distance measuring instruments (laser meter, EDM, etc.) • Indirect distance measuring instruments (level, theodolite, GPS/GNSS receiver, etc.) * Description of surveying tools and instruments used in distance measurement • Uses of distance measuring tools and instruments • Accuracy of distance measuring instruments	

Resources required for the learning outcome	
Equipment	Total station, dump level, theodolite, GPS/GNSS receiver.
Materials	Paints, nails, pegs, beacons, notebook.
Tools	Tape measure, plumb bob, staff reading, hummer, arrows, rod.
Facilitation techniques	• Demonstration and simulation • Individual and group work • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment

Identify levelling tools and equipment		Learning hours: 6
Indicative content		
* Tools used in levelling tools (Refer to ISO 17123-2, ISO 1141) • Rods, plumb bob, hammers, measuring tapes, arrows and pins * Instruments used in levelling • Level, theodolite * Description of levelling tools and instruments • Uses of levelling tools and instruments • Accuracy levelling instruments		
Resources required for the indicative content		
Equipment	Total station, level, theodolite.	
Materials	Paints, nails, pegs, etc.	
Tools	Plumb bob, staff reading, Prism square, hummer, ranging rod arrows and pins.	
Facilitation techniques	• Demonstration and simulation • Individual and group work • Trainer guided • Group discussion	

Identify angular measuring tools and equipment		Learning hours:8
Indicative content		
Tools used in angular measurement (Refer to ISO 8322-4, ISO 17123-3) Arrows, ranging rods, protractor, square		
* Instrument used in angular measurement - Compass, Theodolite, Level, total station		
* Description of angular measuring tools and instruments - Uses of angular measuring tools and instruments - Accuracy of angular measuring instruments		
Resources required for the indicative content		
Equipment	Theodolite , total station, level, etc.	
Materials	Paints, pegs, beacons.	
Tools	Plumb bob, staff reading, prism square, hummer, protractor, rod, arrows.	
Facilitation techniques	- Demonstration and simulation - Individual and group work - Trainer guided - Group discussion	
Formative assessment methods	- Written assessment - Oral assessment	

Identify coordinate recording and tracking instrument		Learning hours:8
Indicative content		
* Types of surveying instruments used in coordinate tracking (Refer to ISO 17123-5, ISO 17123-8:2015) - Satellite-based instrument (GNSS, radar), - Ground-based instrument (Total station, laser scanner etc.) - Aerial base instrument (Drones, etc.)		
* Description of satellite-based coordinate tracking instrument - Uses of satellite-based coordinate tracking instruments - Accuracy of satellite-based coordinate tracking instruments		
* Description of ground-based coordinate tracking instrument - Uses of ground-based coordinate tracking instruments		

• Accuracy of ground-based coordinate tracking instruments	
Resources required for the indicative content	
Equipment	GNSS, total station, laser scanner.
Materials	Paints, pegs, beacons, etc.
Tools	Plumb bob, prism square, hummer, arrows.
Facilitation techniques	• Demonstration and simulation • Individual and group work • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment

References:

1. Viner, D (2015). *Occupational Risk Control: Predicting and Preventing the Unwanted*.
2. OSHA General Industry regulations Book, 29 CFR 1910 Paperback- July1,2014 y MANCOMM Inc. (Author, Editor)
3. Archer, R and Borthwick, K.(2009).*OH&S: a management guide*.
4. G:\\Basic OH&S Program Elements OSH Answers.htm
5. <https://www.rospace.com/occupational-safety/advice/training-matters/>
6. <https://www.3tonline.fi/incident-reporting>
7. <http://www.safety.uwa.edu.au/topics/plant/inspection>
8. <http://www.hse.gov.uk/statistics/causdis/index.htm>
9. <https://www.ccohs.ca/oshanswers/hsprograms/job-haz.html>.
10. ISO 8322-7 Building construction — Measuring instruments — Procedures for determining accuracy in use — Part 7: Instruments when used for setting out
11. ISO 17123-2 Optics and optical instruments — Field procedures for testing geodetic and surveying instruments — Part 2: Levels
12. ISO 1141 Fibre Ropes — Polyester — 3-, 4-, 8- And 12-Strand Ropes
13. ISO 8322-4 Building construction — measuring instruments — Procedures for determining accuracy in use — Part 4: Theodolites
14. ISO 17123-2:2001 Optics and optical instruments — Field procedures for testing geodetic and surveying instruments — Part 2: Levels
15. ISO 17123-3:2001 Part 3: Theodolites
16. ISO 17123-5:2018 Part 5: Total stations
17. ISO 17123-8:2015 Part 8: GNSS field measurement systems in real-time kinematic (RTK)